

```

1  #include <stdio.h>
2  #include <string.h>
3
4  #define MAX_DIR 10
5  #define MAX_FILES 20
6
7  // File structure
8+ struct file {
9      char fname[20];
10 };
11
12 // Directory structure
13+ struct directory {
14     char dname[20];
15     struct file files[MAX_FILES];
16     int fileCount;
17 };
18
19 struct directory dir[MAX_DIR];
20 int dirCount = 0;
21
22 // Create directory
23+ void createDirectory() {
24+     if (dirCount >= MAX_DIR) {
25         printf("Directory limit reached!\n");
26         return;
27     }
28
29     printf("Enter directory name: ");
30     scanf("%s", dir[dirCount].dname);
31
32     dir[dirCount].fileCount = 0;
33     dirCount++;
34
35     printf("Directory created successfully.\n");
36 }
37
38 // Create file inside a directory
39+ void createFile() {
40     char dname[20], fname[20];
41     int found = 0;
42
43     printf("Enter directory name: ");
44     scanf("%s", dname);
45
46+     for (int i = 0; i < dirCount; i++) {
47+         if (strcmp(dir[i].dname, dname) == 0) {
48             found = 1;
49
50+             if (dir[i].fileCount >= MAX_FILES) {

```

```

50-         if (dir[i].fileCount >= MAX_FILES) {
51             printf("File limit reached!\n");
52             return;
53         }
54
55         printf("Enter file name: ");
56         scanf("%s", fname);
57
58         strcpy(dir[i].files[dir[i].fileCount].fname, fname);
59         dir[i].fileCount++;
60
61         printf("File created successfully.\n");
62         break;
63     }
64 }
65
66-     if (!found) {
67         printf("Directory not found!\n");
68     }
69 }
70
71 // List all directories
72- void listDirectories() {
73-     if (dirCount == 0) {
74         printf("No directories available.\n");
75         return;
76     }
77
78     printf("Directories:\n");
79-     for (int i = 0; i < dirCount; i++) {
80         printf("%s\n", dir[i].dname);
81     }
82 }
83
84 // List files in a directory
85- void listFiles() {
86     char dname[20];
87     int found = 0;
88
89     printf("Enter directory name: ");
90     scanf("%s", dname);
91
92-     for (int i = 0; i < dirCount; i++) {
93-         if (strcmp(dir[i].dname, dname) == 0) {
94             found = 1;
95
96-             if (dir[i].fileCount == 0) {
97                 printf("No files in this directory.\n");
98                 return;
99             }
100

```

```

101         printf("Files in %s:\n", dname);
102         for (int j = 0; j < dir[i].fileCount; j++) {
103             printf("%s\n", dir[i].files[j].fname);
104         }
105         break;
106     }
107 }
108
109 if (!found) {
110     printf("Directory not found!\n");
111 }
112 }
113
114 int main() {
115     int choice;
116
117     while (1) {
118         printf("\n1. Create Directory\n2. Create File\n3. List Directories\n4. List Files\n5. Exit\n");
119         printf("Enter your choice: ");
120         scanf("%d", &choice);
121
122         switch (choice) {
123             case 1:
124                 createDirectory();
125                 break;
126             case 2:
127                 createFile();
128                 break;
129             case 3:
130                 listDirectories();
131                 break;
132             case 4:
133                 listFiles();
134                 break;
135             case 5:
136                 return 0;
137             default:
138                 printf("Invalid choice!\n");
139         }
140     }
141 }

```

```

1. Create Directory
2. Create File
3. List Directories
4. List Files
5. Exit
Enter your choice: 1
Enter directory name: os
Directory created successfully.

```

```

1. Create Directory
2. Create File
3. List Directories
4. List Files
5. Exit
Enter your choice: 2
Enter directory name: file
Directory not found!

```

```

1. Create Directory
2. Create File
3. List Directories
4. List Files
5. Exit
Enter your choice:

```